

Earnout Settlement Ledger

The contingent part of an acquisition price stops being a promise only the buyer can score, and becomes a computation both sides sign at closing and the ledger executes every quarter.

Built on Canton · Daml model 0.0.6 · Bonus challenge: BitSafe (decentralization)

THE PROBLEM

An earnout is the part of the price the buyer gets to grade

The deferred, performance-linked portion of an acquisition price — typically about a third of the closing price, paid out over two to four years if the acquired business hits agreed targets.

24%

of private-target acquisition agreements
contain an earnout

ABA Private Target Deal Points Study

28%

of earnouts end in dispute

SRS Acquiom

21¢

collected per promised dollar of
contingent consideration

SRS Acquiom, Kroll

Disputes resolve through accounting arbitration or the Delaware Court of Chancery: one to three years, six-figure fees, starting from numbers only one side produced.

At closing, control of the evidence transfers to one side

The buyer now owns the ERP, the chart of accounts, the cost allocation keys, and the working definition of recognized revenue. Every quarter the buyer computes the metrics and announces whether the target was met.

**The seller cannot verify that number. An honest buyer cannot prove it.
Good faith becomes unfalsifiable in both directions.**

This is structural, not behavioural. The second-order cost is larger than the litigation: because every experienced dealmaker knows the contingent portion is unreliable, buyers promise less and sellers believe less. The gap the earnout was invented to bridge stays open.

Every existing fix fires after the harm

WHAT DEAL TEAMS TRY TODAY	WHY IT DOES NOT CLOSE THE GAP
Drafting harder EBITDA defined line by line, accounting policies frozen at closing	A clause is only as strong as the forum that enforces it — and the forum is chosen after the dispute has already started.
Contractual audit rights	An audit inspects the <i>output</i> of the buyer's systems, months after the period closed. Never the computation.
Escrow, minimums, acceleration	Padding lowers the exposure and the headline price — exactly what the seller was trying to avoid.
Expert determination a Big Four firm named as tiebreaker	Still starts from numbers only one side produced, so it collapses into an argument about credibility rather than arithmetic.

The enforcement has to move from after the dispute to before the signature.

THE SOLUTION

Lock the rules at signing. Let the ledger run them.

BUYER + SELLER

1 • Co-sign

Metric definitions, thresholds, evaluation logic, attester and arbiter — signed once. Both signatures are required for the contract to exist.

BANK

2 • Attest

The acquired entity's bank signs the period's account snapshot on-ledger. Facts enter from a party with no stake in the payout.

BUYER

3 • Report

Period metrics submitted with hashes of the supporting evidence. The hash binds the claim to a document set without exposing it.

CONTRACT

4 • Evaluate

The agreed logic runs itself and records the earned amount. Anomalies route to the arbiter named at signing.

CONTRACT

5 • Settle

The final payout transfers in a single atomic transaction. No "judgment won, payment stalled" gap.

There is no function to change the rules afterwards. The attempt cannot even be expressed as a transaction.

THE DESIGN

Four parties, because responsibility is decomposed

Facts come from the attestor, rules from the contract, interpretation from the arbiter. No single party holds the ledger.

Buyer

Proposes terms, reports each period against the attested snapshot. Cannot amend what was signed.

Seller

Co-signs the terms, then verifies every quarterly result without an audit.

Attestor (bank)

Signs the account snapshot each period. Sees the snapshot, not the earnout economics.

Arbiter

Resolves flagged quarters only. Scope fixed at signing; reasoning recorded on-ledger.

A three-party design would leave the buyer as the sole source of the underlying facts — which is the exact defect this product exists to remove.

The transparent-chain test: what breaks if this moves to a public chain tomorrow?

- **The acquirer's quarterly financials would be public.** A private company's revenue, cash flow and account activity cannot go on a transparent chain. Here the seller sees verdicts and metric values only; raw transaction data stays off-chain, hash-committed; the arbiter reaches the basis only on dispute.
- **A central database has no home.** The buyer's system? The seller will not trust it — that *is* the problem. The seller's? Symmetric. An escrow agent holds money but does not adjudicate. The product is precisely a record neither side can own or amend.
- **Adjudication and payment settle atomically.** Verdict finalization creates the payment obligation; the transfer and the contract's closure happen in one transaction.

Deal terms are confidential. An earnout published to a transparent ledger is not adoptable by any real deal team, regardless of how correct the logic is. Sub-transaction privacy is the precondition — which is why this is a Canton application rather than a chain-agnostic one.

PROOF

Seven attacks, every one stopped by the ledger itself

ATTACK	RESULT ON THE LEDGER
Buyer creates a favorable verdict alone	rejected — both signatures required
Buyer self-issues a bank attestation	rejected — attestor signature required
Reusing a consumed attestation	rejected — single-use, archived on submission
Resubmitting or skipping a quarter	rejected — agreement tracks nextQuarter
Padding finalization with duplicate verdicts	rejected — deal-scoped, distinct-quarter check
Reporting metrics with no bank attestation	impossible — the choice takes an attestation contract
Amending terms after signing	impossible — no such choice exists

Run them yourself: `dpm test --package-root tests` — 21 transactions, **5 asserted must-fail scenarios**. The last two attacks need no test: the model cannot express them. The demo shows the attack as a **failing transaction on screen**, not as a claim on a slide.

STATUS

What exists today, and what lands during Season 3

Built · pre-hackathon, disclosed

- Daml model 0.0.6, four-party workflow, end to end on LocalNet (JSON Ledger API v2, OAuth2/Keycloak).
- Five negative tests: self-issued attestation, reused attestation, out-of-order quarter, solo verdict, padded finalization — all asserted to fail.
- Three-panel operator UI, KR/EN. 59-second captioned demo.

Season 3 · Sep 18 – Oct 9

- **Clause replay harness, and five public clauses through it** (7b8736d, 5e9bf38, f566744). A clause compiles into the signed terms or is refused with the reason. Prosser v. PharmaLogic replayed to the \$0 actually paid; four refused, reasons published.
- **A clause self-check** (96a5356) — the rules read a clause's shape, not its figures, so they run in the reader's own browser. One line comes back, and that line is the finding.
- Still open: **DevNet deployment**, per-party users.

Every commit from Sep 18 is in docs/s3-log.md with its hash, and a fresh clone yields a byte-identical DAR (sha256 0390e7c1...). None of this needs taking on trust.

What we are not claiming

- **Zero user interviews so far.** The audience definition is a hypothesis built on secondary data, not primary research. Anyone claiming a validated audience three weeks into a hackathon is selling you something.
- **The contract enforces the computation, not the truthfulness of the inputs.** A buyer who misreports revenue is still misreporting. What changes is that the dispute narrows from "what did the clause mean" to "is this one figure correct" — answerable by an arbiter in weeks rather than by a court in years.
- **Evidence hashes bind a claim to a document set.** They do not prove the documents are right.

Our success metric is deliberately the one that could kill the project — and it already fired: **clauses we did not write must compile into the signed terms and replay to the payout that was actually made.** Five public-record clauses are in. One, Prosser v. PharmaLogic, replays to the \$0 actually paid. Four are refused as inexpressible, with the reason published for each.

THE ASK

Introductions to people who have lived through a broken earnout

Sell-side M&A advisors, deal lawyers, and escrow or paying agents who will replay one closed earnout against our evaluation logic using their own numbers.

A replay that reproduces a dispute they lived through is worth more than any deck, including this one.